



university of
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Models and Semantics of Computation (MaSC)

Lecture 3: Well-Founded Induction and IMP

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From Lecture 1

Suppose that we have expressions with **variables**, denoted $x, y, \dots$:

$$E ::= x \mid N \mid E \oplus E \mid E \otimes E$$

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$$\langle E, \sigma \rangle \qquad \mathcal{E}[\![\cdot]\!] : Exp \rightarrow (\mathbb{M} \rightarrow \mathbb{N})$$

- ▶ How to redefine the various semantics and properties?

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- ▶ How to redefine the various semantics and properties?

Today: A proof technique (well-founded induction) and the syntax of IMP



Part I

Motivation



- How to prove an **existential statement**?

$$\exists x. P(x)$$

Proof Techniques



- ▶ How to prove an **existential statement**?
→ Exhibit a **witness**.

$$\exists x. P(x)$$

Proof Techniques



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$\exists n \in \mathbb{N}. n^2 \leq n$	$n = 0$

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Statement	Counterexample
$\forall n \in \mathbb{N}. n^2 \leq n$	$n = 2$

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- ▶ How to disprove a **universal statement**?

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- ▶ Prove a **universal statement**?

→ Use **induction**!

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Today: Well-founded induction
Induction on well-founded relations

What is Common To



- ▶ natural numbers
- ▶ lists
- ▶ trees
- ▶ grammar languages
- ▶ terms of a signature
- ▶ theorems of a logic system
- ▶ derivations
- ▶ computations

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base cases
inductive cases

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	base case	inductive case
natural numbers		
lists		
trees		
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What is Common To



	base case	inductive case
natural numbers	0	succ
lists		
trees		
grammar languages		
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What is Common To



	base case	inductive case
natural numbers	0	succ
lists	nil	cons
trees		
grammar languages		
terms of a signature		
theorems of a logic system		
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computations		

What is Common To



	base case	inductive case
natural numbers	0	succ
lists	nil	cons
trees	nil	node
grammar languages		
terms of a signature		
theorems of a logic system		
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What is Common To



	base case	inductive case
natural numbers	0	succ
lists	nil	cons
trees	nil	node
grammar languages	productions with terminal symbols only	productions with non-terminal symbols
terms of a signature		
theorems of a logic system		
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computations		

What is Common To



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natural numbers	0	succ
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trees	nil	node
grammar languages	productions with terminal symbols only	productions with non-terminal symbols
terms of a signature	constants	operators
theorems of a logic system		
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theorems of a logic system	axioms	inference rules
derivations	axioms	inference rules
computations	single step	concatenation



Part II

Well-Founded Induction



Ingredients

- ▶ A set of elements A , possibly infinite.
- ▶ A predicate $P : A \rightarrow \mathbb{B}$. We want to prove $\forall a \in A. P(a)$.
- ▶ A binary relation $\prec \subseteq A \times A$, not necessarily transitive.



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 - ▶ $a \prec b$ reads ' a precedes b '
 - ▶ also written $b \succ a$
 - ▶ also written $a \rightarrow b$ (graph notation)



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- ▶ To use induction, we must guarantee to reach some base cases.



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 - ▶ also written $b \succ a$
 - ▶ also written $a \rightarrow b$ (graph notation)
- ▶ To use induction, we must guarantee to reach some base cases.
Hence, no infinite descending chain is allowed in $\prec$.
That is, $\prec$ must be well-founded.

Well-founded Induction Principle



Well-founded Induction

Let $\prec \subseteq A \times A$ be well-founded.

$$(\forall a \in A. P(a)) \Leftrightarrow (\forall a \in A. (\forall b \prec a. P(b)) \Rightarrow P(a))$$

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Key ideas in the proof of the principle:

- ▶ A relation is well-founded iff its **transitive closure** is well-founded
- ▶ Well-founded relations are **acyclic**
- ▶ $\prec$ is well-founded iff any $Q \subseteq A$ has a **minimal element**

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Goal: Derive **useful instances** of the induction principle, to reason about IMP.

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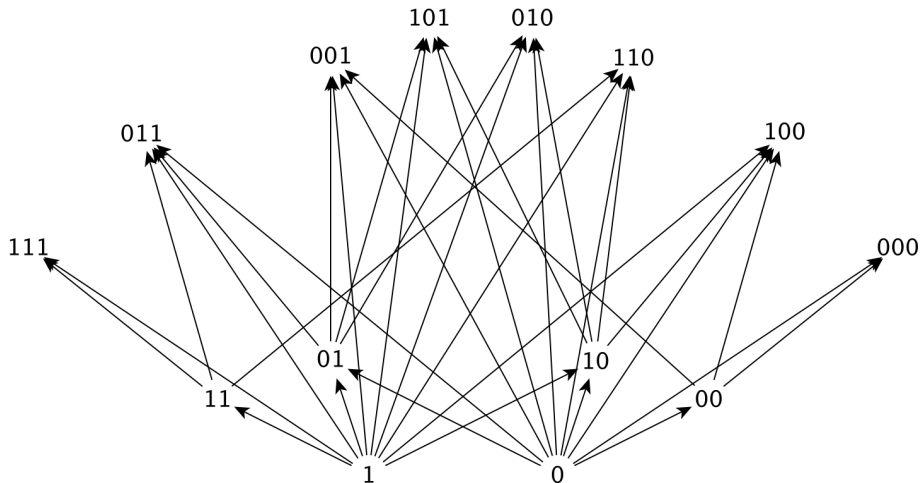
Recall:

- $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$ (this is the **contrapositive formulation**).

Example: Graph of a Relation



$A = \mathbb{B}^*$, with $u \prec w$ if u appears in w (with $u \neq \epsilon$ and $u \neq w$)



Infinite Descending Chain



An infinite sequence $\{a_i\}_{i \in \mathbb{N}}$ of elements in A
such that $\forall i \in \mathbb{N}. a_i \succ a_{i+1}$

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The sequence can also be seen as a function $a : \mathbb{N} \rightarrow A$, such that $a(i)$ decreases (in the sense of $\prec$) as i grows:

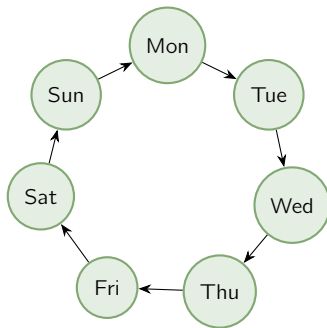
$$a(0) \succ a(1) \succ a(2) \succ \dots$$

Infinite Descending Chain



Example

- ▶ $A = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\}.$
- ▶ $\text{Sat} \prec \text{Sun} \prec \text{Mon} \prec \dots$
(equivalently: $\text{Mon} \succ \text{Sun} \succ \text{Sat} \succ \dots$)
- ▶ $a(n) = n\text{th day past}$



Well-founded Relations



A relation is called **well-founded**
if has no infinite descending chain.

		well-founded?
$\mathbb{N}$	$n \prec m$ if $m = n + 1$	
$\mathbb{Z}$	$n \prec m$ if $m = n + 1$	
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$\mathbb{N}$	$n \prec m$ if $n \leq m$	
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In general, a well-founded relation cannot be reflexive.

Transitive Closure



Given a relation $\prec$, its **transitive closure** $\prec^+$ is the least relation generated by the following rules:

$$\frac{a \prec b}{a \prec^+ b}$$

$$\frac{a \prec^+ b \quad b \prec^+ c}{a \prec^+ c}$$

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From this definition:

$$\begin{aligned} \prec &\subseteq \prec^+ \\ (\prec^+)^+ &= \prec^+ \end{aligned}$$

Transitive and Reflexive Closure



Given a relation $\prec$, its **transitive and reflexive closure** $\prec^*$ is the least relation generated by the following rules:

$$\frac{a \in A}{a \prec^* a}$$

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From this definition:

$$\begin{aligned} \prec &\subseteq \prec^+ \subseteq \prec^* \\ (\prec^*)^* &= \prec^* \end{aligned}$$

Closures and Induced Paths



Given $\prec$, we have:

$a \prec^+ b$ iff there is a **non-empty**, finite path from a to b in the graph of $\prec$
 $\exists k > 0, \{c_i\}_{i \in [0, k]}. a = c_0 \prec c_1 \prec \dots \prec c_k = b$

$a \prec^* b$ iff there is a **possibly empty**, finite path from a to b in the graph of $\prec$
 $\exists k \geq 0, \{c_i\}_{i \in [0, k]}. a = c_0 \prec c_1 \prec \dots \prec c_k = b$

Closures, By Example



	$\prec^+$	$\prec^*$
$\mathbb{N}$ $n \prec m$ if $m = n + 1$	$n < m$	$n \leq m$
$\mathbb{Z}$ $n \prec m$ if $m = n + 1$	$n < m$	$n \leq m$
$\mathbb{N}$ $n \prec m$ if $n < m$	$n < m$	$n \leq m$
$\mathbb{N}$ $n \prec m$ if $n \leq m$	$n \leq m$	$n \leq m$
$\mathbb{N}$ $n \prec m$ if $n = m$	$n = m$	$n = m$

Theorem 4.2



A relation $\prec$ is well-founded iff its transitive closure $\prec^+$ is well-founded.

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There are two directions:

1. $\prec^+ \text{ w.f.} \Rightarrow \prec \text{ w.f.}$

2. $\prec \text{ w.f.} \Rightarrow \prec^+ \text{ w.f.}$

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1. $\prec^+ \text{ w.f.} \Rightarrow \prec \text{ w.f.}$

Any descending chain for $\prec$ is a descending chain for $\prec^+$. By assumption, the descending chains for $\prec^+$ are finite, so are the descending chains for $\prec$.

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Any descending chain for $\prec$ is a descending chain for $\prec^+$. By assumption, the descending chains for $\prec^+$ are finite, so are the descending chains for $\prec$.

2. $\prec \text{ w.f.} \Rightarrow \prec^+ \text{ w.f.} \equiv \neg(\prec^+ \text{ w.f.}) \Rightarrow \neg(\prec \text{ w.f.})$

Consider an infinite descending chain for $\prec^+$:

$$a_0 \succ^+ a_1 \succ^+ a_2 \succ^+ \dots$$

Recall that $a \succ^+ b$ iff there is a non-empty, finite path from a to b in the graph of $\prec$. Therefore, we derive the infinite descending chain in $\prec$:

$$a_0 \succ \dots \succ a_1 \succ \dots \succ a_2 \succ \dots \succ \dots$$

Acyclic Relations



- ▶ We say that the relation $\prec$ has a cycle if $a \prec^+ a$, for some $a \in A$.
- ▶ We say that $\prec$ is **acyclic** if it has no cycles: $\forall a \in A. a \not\prec^+ a$.
- ▶ Note: $\prec$ is acyclic iff $\prec^+$ is acyclic.

Theorem 4.3



If $\prec$ is well-founded then it is acyclic.

By contraposition:

Theorem 4.3



If $\prec$ is well-founded then it is acyclic.

By contraposition: we prove that if $\prec$ has a cycle then it is not well-founded.

- Take a $a \in A$ such that $a \prec^+ a$. We build an infinite descending chain:

$$a \succ^+ a \succ^+ a \succ^+ \dots$$

- Hence, $\succ^+$ is not well-founded. By Theorem 4.2, then $\succ$ is not well-founded.

Minimal Element



Let $\prec$ be a relation over A .

- ▶ Given $Q \subseteq A$, we say $m \in Q$ is **minimal** (wrt $\prec$) if there is no $x \in Q$ such that $x \prec m$. That is, $\forall x \in Q. m \prec x$.
- ▶ Q has no minimal element if $\forall m \in Q. \exists x \in Q. x \prec m$.

Lemma 4.1



$\prec$ is well-founded iff
every nonempty $Q \subseteq A$ contains a minimal element m .

Lemma 4.1



- (1) $\prec$ has an infinite descending chain iff
- (2) there is a non-empty $Q \subseteq A$ with no minimal element

Lemma 4.1



- (1) $\prec$ has an infinite descending chain iff
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► (1) $\Rightarrow$ (2):

Take an infinite descending chain $a_1 \succ a_2 \succ a_3 \succ \dots$ and consider the associated set $Q = \{a_1, a_2, a_3, \dots\}$. The set Q has no minimal element: for every $a_i \in Q$ we know that there is a $a_{i+1} \in Q$ such that $a_i \succ a_{i+1}$.

Lemma 4.1



- (1) $\prec$ has an infinite descending chain iff
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Take an infinite descending chain $a_1 \succ a_2 \succ a_3 \succ \dots$ and consider the associated set $Q = \{a_1, a_2, a_3, \dots\}$. The set Q has no minimal element: for every $a_i \in Q$ we know that there is a $a_{i+1} \in Q$ such that $a_i \succ a_{i+1}$.

► (2) $\Rightarrow$ (1):

We consider a non-empty set $Q \subseteq A$ with no minimal element.

Take any $a_0 \in Q$: because it is not minimal, there is a a_1 such that $a_0 \succ a_1$. By a similar reasoning, a_1 is not minimal either; we construct an infinite descending chain by iterating the argument.

Theorem 4.5



Let $\prec$ be a well-founded relation over A .

$$\underbrace{\forall a \in A. P(a)}_{(1)} \Leftrightarrow \underbrace{(\forall a \in A. (\forall b \prec a. P(b)) \Rightarrow P(a))}_{(2)}$$



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Set $H(a) \triangleq \forall b \prec a. P(b)$ and $S(a) \triangleq H(a) \Rightarrow P(a)$.



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► **(1) $\Rightarrow$ (2):**

Take an arbitrary $a \in A$. By assumption, $P(a)$. We have:

$$\begin{aligned} S(a) &\equiv H(a) \Rightarrow P(a) \\ &\equiv (\neg H(a) \vee P(a)) \\ &\equiv (\neg H(a) \vee \text{true}) \\ &\equiv \text{true} \end{aligned}$$

Theorem 4.5



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Set $H(a) \triangleq \forall b \prec a. P(b)$ and $S(a) \triangleq H(a) \Rightarrow P(a)$.

► **(2) $\Rightarrow$ (1):** (This is the direction we really want!)

We prove $\neg(1) \Rightarrow \neg(2)$. Assume $\exists a. \neg P(a)$.

Let $Q = \{q \in A \mid \neg P(q)\} \neq \emptyset$.

Since $\prec$ is well-founded, then Q has a minimal element $m \in Q$ (Lemma 4.1).

Clearly, $\neg P(m)$. Because m is minimal, we have $\forall b \prec m. P(b) \equiv H(m)$.

Thus, $H(m) \wedge \neg P(m) \equiv \neg(H(m) \Rightarrow P(m)) \equiv \neg S(m)$.

Therefore, $\exists a \in A. \neg S(a)$.

Well-Founded Induction



Let $\prec \subseteq A \times A$ be a well-founded relation.

$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

- A general **proof principle**, aka Noetherian induction.

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- ▶ A **base case** is any element of A such that the set $\{b \in A \mid b \prec a\}$ is empty.

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We may now **instantiate the principle**, by choosing specific A and $\prec$.



Part III

Instances

Instance 1: Mathematical Induction



- ▶ Set: $A = \mathbb{N}$
- ▶ Well-founded relation: $\prec = \{(n, n + 1) \mid n \in \mathbb{N}\}$ (immediate precedence)

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$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

Two cases:

1. Case $a = 0$: There is no $b \prec 0$ and so $(\forall b \prec 0. P(b)) \equiv \text{true}$ and

$$((\forall b \prec 0. P(b)) \Rightarrow P(0)) \equiv \text{true} \Rightarrow P(0)$$

$$\equiv \boxed{P(0)}$$

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2. Case $a = n + 1$: There is only one b such that $b \prec n + 1$, i.e., $b = 1$, and

$$((\forall b \prec n + 1. P(b)) \Rightarrow P(n + 1)) \equiv \boxed{P(n) \Rightarrow P(n + 1)}$$

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$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

$\rightsquigarrow$

$$\frac{P(0) \quad \forall n \in \mathbb{N}. (P(n) \Rightarrow P(n + 1))}{\forall a \in A. P(a)}$$

Instance 2: Strong Induction



- Set: $A = \mathbb{N}$
- Well-founded relation: $\prec = <$

(strictly-less than)

$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

Two cases:

1. Case $a = 0$: As before, there is no $b \prec 0$ and so $(\forall b \prec 0. P(b)) \equiv \text{true}$ and

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$$((\forall b \prec 0. P(b)) \Rightarrow P(0)) \equiv \boxed{P(0)}$$

2. Case $a = n + 1$: There are multiple b such that $b \prec n + 1$ and

$$((\forall b \prec n + 1. P(b)) \Rightarrow P(n + 1)) \equiv \boxed{P(0) \wedge \cdots \wedge P(n) \Rightarrow P(n + 1)}$$

Instance 2: Strong Induction



- ▶ Set: $A = \mathbb{N}$
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$$\frac{P(0) \quad \forall n \in \mathbb{N}. ((P(0) \wedge \dots \wedge P(n)) \Rightarrow P(n+1))}{\forall a \in A. P(a)}$$

Instance 3: Structural Induction



- ▶ Consider a signature $\Sigma = \{\Sigma_n\}_{n \in \mathbb{N}}$.
- ▶ Set $A = T_\Sigma$ (closed terms)
- ▶ Define the **immediate subterm relation** $\prec$:

$$\prec = \left\{ (t_i, f(t_1, \dots, t_n)) \mid f \in \Sigma_n, i \in [1..n] \right\}$$

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Example

Let $\Sigma_0 = \{0\}$, $\Sigma_1 = \{\text{succ}\}$, and $\Sigma_2 = \{\text{plus}\}$. We have:

- ▶ $0 \prec \text{succ}(0) \prec \text{plus}(0, \text{succ}(0))$

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- ▶ $0 \not\prec \text{plus}(\text{succ}(0), \text{succ}(0))$

Before instantiating the principle, we need to prove that $\prec$ is well founded.

Subterm Relation is Well Founded



In the proof, we relate $\prec$ to a known well-founded relation:

- Let $depth : T_{\Sigma} \rightarrow \mathbb{N}$ be defined as

$$\begin{aligned} depth(c) &\triangleq 1 && \text{if } c \in \Sigma_0 \\ depth(f(t_1, \dots, t_n)) &\triangleq 1 + \max_{i \in [1..n]} depth(t_i) && \text{if } f \in \Sigma_n \end{aligned}$$

- By definition, if $t \prec t'$ then $depth(t) < depth(t')$.
- Any descending chain in $\prec$ induces a descending chain in $<$.
- Since $<$ is well-founded, so is $\prec$.

Corollary



- Because $\prec$ is well-founded, its transitive closure $\prec^+$ is well-founded.

Example

Let $\Sigma_0 = \{0\}$, $\Sigma_1 = \{\text{succ}\}$, and $\Sigma_2 = \{\text{plus}\}$. We have:

- $0 \prec^+ \text{succ}(0) \prec^+ \text{plus}(0, \text{succ}(0))$
- $0 \prec^+ \text{plus}(0, \text{succ}(0))$
- $0 \prec^+ \text{plus}(\text{succ}(0), \text{succ}(0))$

Instance 3: Structural Induction



- ▶ Set: $A = T_\Sigma$ (closed terms)
- ▶ Well-founded relation: $\prec = \{(t_i, f(t_1, \dots, t_n)) \mid f \in \Sigma_n, i \in [1..n]\}$

$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

$\rightsquigarrow$

$$\frac{\forall n \in \mathbb{N}. \forall f \in \Sigma_n. \forall t_1, \dots, t_n. (P(t_1) \wedge \dots \wedge P(t_n)) \Rightarrow P(f(t_1, \dots, t_n))}{\forall t \in T_\Sigma. P(t)}$$



Part IV

Induction At Work

IMP: A Language in Three Layers

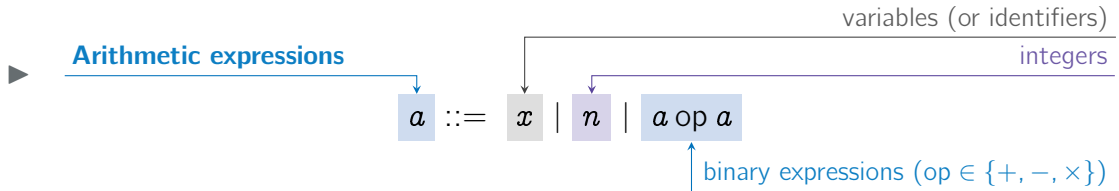


Arithmetic expressions



$a ::= x \mid n \mid a \text{ op } a$

IMP: A Language in Three Layers



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Arithmetic expressions

$a ::= x \mid n \mid a \text{ op } a$

Boolean expressions

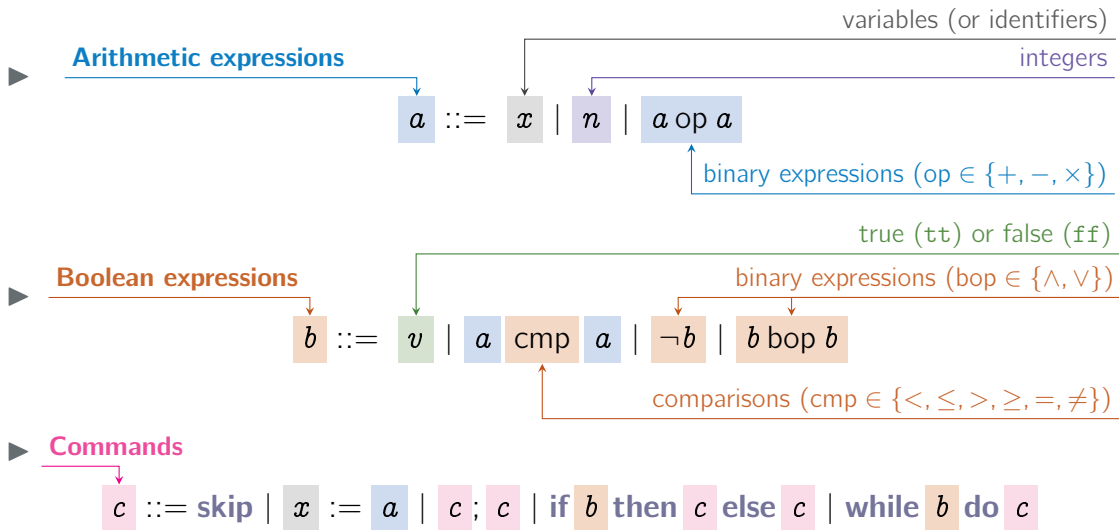
$b ::= v \mid a \text{ cmp } a \mid \neg b \mid b \text{ bop } b$

true (tt) or false (ff)

binary expressions (bop $\in \{\wedge, \vee\}$)

comparisons (cmp $\in \{<, \leq, >, \geq, =, \neq\}$)

IMP: A Language in Three Layers



IMP: A Language in Three Layers



The syntax of IMP:

$$a ::= x \mid n \mid a \text{ op } a$$
$$b ::= v \mid a \text{ cmp } a \mid \neg b \mid b \text{ bop } b$$
$$c ::= \text{skip} \mid x := a \mid c; c \mid \text{if } b \text{ then } c \text{ else } c \mid \text{while } b \text{ do } c$$

Let's focus for the moment on arithmetic expressions and their properties.



Syntax

$x \in \text{Ide}$ $n \in \mathbb{Z}$ $\mathbb{M} \triangleq \{\sigma \mid \text{Ide} \rightarrow \mathbb{Z}\}$ $\text{op} \in \{+, -, \times\}$
 $a ::= x \mid n \mid a \text{ op } a$

Semantics



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$x \in \text{Ide}$ $n \in \mathbb{Z}$ $\mathbb{M} \triangleq \{\sigma \mid \text{Ide} \rightarrow \mathbb{Z}\}$ $\text{op} \in \{+, -, \times\}$
 $a ::= x \mid n \mid a \text{ op } a$

Semantics

$\overline{\langle x, \sigma \rangle} \longrightarrow \sigma(x)$



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$$x \in \text{Ide} \quad n \in \mathbb{Z} \quad \mathbb{M} \triangleq \{\sigma \mid \text{Ide} \rightarrow \mathbb{Z}\} \quad \text{op} \in \{+, -, \times\}$$
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Semantics

$$\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)} \quad \frac{}{\langle n, \sigma \rangle \longrightarrow n}$$



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 $a ::= x \mid n \mid a \text{ op } a$

Semantics

$$\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)} \quad \frac{}{\langle n, \sigma \rangle \longrightarrow n} \quad \frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$



Syntax

$$x \in \text{Ide} \quad n \in \mathbb{Z} \quad \mathbb{M} \triangleq \{\sigma \mid \text{Ide} \rightarrow \mathbb{Z}\} \quad \text{op} \in \{+, -, \times\}$$
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A **termination property**:

- ▶ $P(a) \triangleq \forall \sigma \in \mathbb{M}. \exists m \in \mathbb{Z}. \langle a, \sigma \rangle \longrightarrow m.$
- ▶ $\forall a. P(a)?$ Structural Induction!



Given the syntax of arithmetic expressions:

$$a ::= x \mid n \mid a \text{ op } a$$

We have that structural induction is as follows:

$$\frac{\forall x \in \text{Ide} \quad \forall n \in \mathbb{Z} \quad \forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)}{\forall a. P(a)}$$

To establish termination we have two base cases, and one inductive case.

Termination: Base Case (1 of 2)



$$\forall x \in \text{Ide. } P(x)$$



Termination: Base Case (1 of 2)



$$\forall x \in \text{Ide}. P(x)$$

Take some arbitrary identifier $x \in \text{Ide}$. We must prove:

$$P(x) \triangleq \forall \sigma. \exists m. \langle x, \sigma \rangle \longrightarrow m$$



Termination: Base Case (1 of 2)



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Take some arbitrary identifier $x \in \text{Ide}$. We must prove:

$$P(x) \triangleq \forall \sigma. \exists m. \langle x, \sigma \rangle \longrightarrow m$$

- Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
Consider the goal $\langle x, \sigma \rangle \longrightarrow m$, where m is the only variable.

Termination: Base Case (1 of 2)



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Take some arbitrary identifier $x \in \text{Ide}$. We must prove:

$$P(x) \triangleq \forall \sigma. \exists m. \langle x, \sigma \rangle \longrightarrow m$$

- ▶ Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
Consider the goal $\langle x, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule $\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)}$ we have $\langle x, \sigma \rangle \longrightarrow m \nwarrow_{[m=\sigma(x)]} \square$

Termination: Base Case (1 of 2)



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Take some arbitrary identifier $x \in \text{Ide}$. We must prove:

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Consider the goal $\langle x, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule $\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)}$ we have $\langle x, \sigma \rangle \longrightarrow m \nwarrow_{[m=\sigma(x)]} \square$
- ▶ We are done by taking $m = \sigma(x)$.

Termination: Base Case (2 of 2)



$$\forall n \in \mathbb{Z}. P(n)$$

We proceed similarly as before: Take some arbitrary $n \in \mathbb{Z}$. We must prove:

$$P(n) \triangleq \forall \sigma. \exists m. \langle n, \sigma \rangle \longrightarrow m$$



Termination: Base Case (2 of 2)



$$\forall n \in \mathbb{Z}. P(n)$$

We proceed similarly as before: Take some arbitrary $n \in \mathbb{Z}$. We must prove:

$$P(n) \triangleq \forall \sigma. \exists m. \langle n, \sigma \rangle \longrightarrow m$$

- Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
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Termination: Base Case (2 of 2)



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Consider the goal $\langle n, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule $\frac{}{\langle n, \sigma \rangle \longrightarrow n}$ we have $\langle n, \sigma \rangle \longrightarrow m \nwarrow_{[m=n]} \square$

Termination: Base Case (2 of 2)



$$\forall n \in \mathbb{Z}. P(n)$$

We proceed similarly as before: Take some arbitrary $n \in \mathbb{Z}$. We must prove:

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Consider the goal $\langle n, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule $\frac{}{\langle n, \sigma \rangle \longrightarrow n}$ we have $\langle n, \sigma \rangle \longrightarrow m \nwarrow_{[m=n]} \square$
- ▶ We are done by taking $m = n$.

Termination: Inductive Case



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- Take some arbitrary expressions a_0, a_1 . We assume:

$$P(a_0) \triangleq \forall \sigma. \exists m_0. \langle a_0, \sigma \rangle \longrightarrow m_0$$

$$P(a_1) \triangleq \forall \sigma. \exists m_1. \langle a_1, \sigma \rangle \longrightarrow m_1$$

We must prove $P(a_0 \text{ op } a_1) \triangleq \forall \sigma. \exists m. \langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$.

Termination: Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$



Termination: Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
Consider the goal $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$, where m is the only variable.

Termination: Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- ▶ Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
Consider the goal $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule
$$\frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$
 we have
$$\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m \nwarrow_{[m=n_0 \text{ op } n_1]} \langle a_0, \sigma \rangle \longrightarrow m_0, \langle a_1, \sigma \rangle \longrightarrow m_1$$

Termination: Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- ▶ Let $\sigma \in \mathbb{M}$ be some arbitrary memory.
Consider the goal $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$, where m is the only variable.
- ▶ By rule
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 we have
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- ▶ By the **inductive hypotheses**, there are m_0, m_1 such that $\langle a_0, \sigma \rangle \longrightarrow m_0$ and $\langle a_1, \sigma \rangle \longrightarrow m_1$

Termination: Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

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 we have
$$\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m \nwarrow_{[m=m_0 \text{ op } m_1]} \langle a_0, \sigma \rangle \longrightarrow m_0, \langle a_1, \sigma \rangle \longrightarrow m_1$$
- ▶ By the **inductive hypotheses**, there are m_0, m_1 such that $\langle a_0, \sigma \rangle \longrightarrow m_0$ and $\langle a_1, \sigma \rangle \longrightarrow m_1$
- ▶ We are done by taking $m = m_0 \text{ op } m_1$.

Another Property of AExpressions



Syntax

$x \in \text{Ide}$ $n \in \mathbb{Z}$ $\mathbb{M} \triangleq \{\sigma \mid \text{Ide} \rightarrow \mathbb{Z}\}$ $\text{op} \in \{+, -, \times\}$
 $a ::= x \mid n \mid a \text{ op } a$

Semantics

$$\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)} \quad \frac{}{\langle n, \sigma \rangle \longrightarrow n} \quad \frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$

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Determinacy:

$$\blacktriangleright P(a) \triangleq \forall \sigma \in \mathbb{M}. \forall m, m' \in \mathbb{Z}. (\langle a, \sigma \rangle \longrightarrow m \wedge \langle a, \sigma \rangle \longrightarrow m') \Rightarrow m = m'.$$

Determinacy: Base Case (1 of 2)



$$\forall x \in \text{Ide. } P(x)$$

Take some arbitrary identifier $x \in \text{Ide}$. We must prove:

$$P(x) \triangleq \forall \sigma, m, m'. \langle x, \sigma \rangle \longrightarrow m \wedge \langle x, \sigma \rangle \longrightarrow m' \Rightarrow m = m'$$

Consider arbitrary σ, m, m' such that $\langle x, \sigma \rangle \longrightarrow m$ and $\langle x, \sigma \rangle \longrightarrow m'$.

We want to prove $m = m'$.

Determinacy: Base Case (1 of 2)



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Take some arbitrary identifier $x \in \text{Ide.}$ We must prove:

$$P(x) \triangleq \forall \sigma, m, m'. \langle x, \sigma \rangle \longrightarrow m \wedge \langle x, \sigma \rangle \longrightarrow m' \Rightarrow m = m'$$

Consider arbitrary σ, m, m' such that $\langle x, \sigma \rangle \longrightarrow m$ and $\langle x, \sigma \rangle \longrightarrow m'$.

We want to prove $m = m'$.

- Consider the goal $\langle x, \sigma \rangle \longrightarrow m$.

Only the rule $\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)}$ is applicable, hence $m = \sigma(x)$.

- Similarly, since $\langle x, \sigma \rangle \longrightarrow m'$, it must be that $m' = \sigma(x)$.

- We thus conclude that $m = m'$.

Determinacy: Base Case (2 of 2)



$$\forall n \in \mathbb{Z}. P(n)$$

Take some arbitrary $n \in \mathbb{Z}$. We must prove:

$$P(n) \triangleq \forall \sigma, m, m'. \langle n, \sigma \rangle \longrightarrow m \wedge \langle n, \sigma \rangle \longrightarrow m' \Rightarrow m = m'$$

Consider arbitrary σ, m, m' such that $\langle n, \sigma \rangle \longrightarrow m$ and $\langle n, \sigma \rangle \longrightarrow m'$.
We want to prove $m = m'$.

- ▶ Consider the goal $\langle n, \sigma \rangle \longrightarrow m$.
Only the rule $\frac{}{\langle n, \sigma \rangle \longrightarrow n}$ is applicable, hence $m = n$.
- ▶ Similarly, since $\langle n, \sigma \rangle \longrightarrow m'$, it must be that $m' = n$.
- ▶ We thus conclude that $m = m'$.

Determinacy: Inductive Case



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

Take arbitrary expressions a_0, a_1 . We assume (inductive hypotheses, $i \in \{0, 1\}$):

$$P(a_i) \triangleq \forall \sigma, m_i, m'_i. \langle a_i, \sigma \rangle \longrightarrow m_i \wedge \langle a_i, \sigma \rangle \longrightarrow m'_i \Rightarrow m_i = m'_i$$

We must prove

$$P(a_0 \text{ op } a_1) \triangleq \forall \sigma, m, m'. (\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m \wedge \langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m') \Rightarrow m = m'$$

Consider generic σ, m, m' such that $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$ and $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m'$.
We want to prove $m = m'$.

Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- Consider the goal $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$.

Only the rule
$$\frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$
 is applicable.

Hence $m = n_0 \text{ op } n_1$ with $\langle a_0, \sigma \rangle \longrightarrow n_0$ and $\langle a_1, \sigma \rangle \longrightarrow n_1$.

- Similarly, since $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m'$, it must be $m' = n'_0 \text{ op } n'_1$ with $\langle a_0, \sigma \rangle \longrightarrow n'_0$ and $\langle a_1, \sigma \rangle \longrightarrow n'_1$.
- By the inductive hypotheses, we have

Inductive Case (cont.)



$$\forall a_0, a_1. P(a_0) \wedge P(a_1) \Rightarrow P(a_0 \text{ op } a_1)$$

- Consider the goal $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m$.

Only the rule
$$\frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$
 is applicable.

Hence $m = n_0 \text{ op } n_1$ with $\langle a_0, \sigma \rangle \longrightarrow n_0$ and $\langle a_1, \sigma \rangle \longrightarrow n_1$.

- Similarly, since $\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow m'$, it must be $m' = n'_0 \text{ op } n'_1$ with $\langle a_0, \sigma \rangle \longrightarrow n'_0$ and $\langle a_1, \sigma \rangle \longrightarrow n'_1$.
- By the inductive hypotheses, we have both $n_0 = n'_0$ and $n_1 = n'_1$.
- We thus conclude $m = n_0 \text{ op } n_1 = n'_0 \text{ op } n'_1 = m'$.

IMP: Big-step Semantics (1 of 2)



$$\frac{}{\langle x, \sigma \rangle \longrightarrow \sigma(x)} \quad \frac{}{\langle n, \sigma \rangle \longrightarrow n} \quad \frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ op } a_1, \sigma \rangle \longrightarrow n_0 \text{ op } n_1}$$

$$\frac{}{\langle v, \sigma \rangle \longrightarrow v} \quad \frac{\langle b, \sigma \rangle \longrightarrow v}{\langle \neg b, \sigma \rangle \longrightarrow \neg v} \quad \frac{\langle a_0, \sigma \rangle \longrightarrow n_0 \quad \langle a_1, \sigma \rangle \longrightarrow n_1}{\langle a_0 \text{ cmp } a_1, \sigma \rangle \longrightarrow n_0 \text{ cmp } n_1}$$

$$\frac{\langle b_0, \sigma \rangle \longrightarrow v_0 \quad \langle b_1, \sigma \rangle \longrightarrow v_1}{\langle b_0 \text{ bop } b_1, \sigma \rangle \longrightarrow v_0 \text{ bop } v_1}$$

IMP: Big-step Semantics (2 of 2)



$$\frac{}{\langle \text{skip}, \sigma \rangle \longrightarrow \sigma} \quad \frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]} \quad \frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'}$$

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff} \quad \langle c_1, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'} \quad \frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c_0, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'}$$

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff}}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma}$$

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$



The End